Analysis of Blood Pressure Levels Among Adults Based on Demographic Factors Using SAS

Sandeep Pudasaini

Spring Semester

2024

Introduction

This research project explores the impact of demographic factors on blood pressure levels among adults, using data from the National Health and Nutrition Examination Survey (NHANES) for the years 2017-2018. The study aims to identify potential areas for targeted public health interventions. Understanding how factors such as age, gender, ethnicity, and socio-economic status influence blood pressure can help in developing more effective health policies and practices. Recent studies have indicated varying impacts of these factors on blood pressure, which this study seeks to further elucidate..

Data Source and Variables

Data Collection: The analysis combines demographic information and blood pressure measurements from the following NHANES datasets:

Demographic Data: NHANES 2017-2018 Demographic Variables and Sample WeightsBlood

Pressure Measurements: NHANES 2017-2018 Blood Pressure and Cholesterol

Variables of Interest:

SEQN: Respondent sequence number (unique identifier)

RIAGENDR: Gender of the respondent

RIDAGEYR: Age in years at the time of screening

RIDRETH3: Race/Hispanic origin, including Non-Hispanic Asian categories

RIDEXMON: Examination month (six-month period of health examination)

DMDEDUC2: Education level for adults aged 20 years and above

INDFMPIR: Income-to-poverty ratio (socioeconomic status)

BPAOCSZ: Coded cuff size used during blood pressure measurement

BPXOSY1: Systolic blood pressure - 1st oscillometric reading

BPXODI1: Diastolic blood pressure - 1st oscillometric reading

Research Question

How do age and income levels influence blood pressure among different gender and ethnic groups?

This question seeks to explore the complex interactions between demographic factors specifically age and income and blood pressure variations across different genders and ethnicities.

Understanding these influences is crucial for identifying key demographic groups that may benefit from targeted public health interventions and preventive healthcare strategies.

Data Preparation and Manipulation

The initial step involved merging the demographic and blood pressure datasets based on the respondent sequence number (SEQN), ensuring a matched pair for each participant. This merger was verified using **PROC CONTENTS**. Missing values were identified and addressed. Specific adjustments included the correction of outliers in blood pressure measurements and alignment of demographic data, verified through consistency checks and statistical summaries post-correction.

Table 1: Merge Dataset

Obs	SEQN	RIAGENDR	RIDAGEYR	RIDRETH3	RIDEXMON	DMDEDUC2	INDFMPIR	BPAOCSZ	BPXOSY1	BPXODI1
1	109264	2	13	1	2	.	0.83	3	109	67
2	109266	2	29	6	2	5	5.00	4	99	56
3	109270	2	11	4	1	.	1.88	3	123	73
4	109271	1	49	3	2	2	.	4	102	65
5	109273	1	36	3	2	4	0.83	3	116	68
6	109274	1	68	7	1	4	1.20	3	138	70
7	109277	2	12	1	1	.	1.35	3	104	58
8	109279	2	17	3	2	.	1.19	3	100	64
9	109282	1	76	3	2	5	3.61	4	141	77
10	109284	2	44	1	1	2	.	.	.	.

Total observations
11656

Step	Total Observations
After removal	6661
Before removal	11656

After removal of missing values dataset

Obs	SEQN	RIAGENDR	RIDAGEYR	RIDRETH3	RIDEXMON	DMDEDUC2	INDFMPIR	BPAOCSZ	BPXOSY1	BPXODI1
1	109266	2	29	6	2	5	5.00	4	99	56
2	109273	1	36	3	2	4	0.83	3	116	68
3	109274	1	68	7	1	4	1.20	3	138	70
4	109282	1	76	3	2	5	3.61	4	141	77
5	109290	2	68	4	2	5	5.00	3	126	62
6	109292	1	58	2	2	3	1.60	3	143	96
7	109293	1	44	3	2	3	0.02	4	126	83
8	109295	2	54	1	1	1	5.00	3	158	96
9	109298	1	68	3	2	1	1.75	4	94	51
10	109300	2	54	6	2	5	5.00	3	155	95

Using Proc SQL

Table 2: Average Blood Pressure by Gender and Race/Ethnicity

Average Blood Pressure by Gender and Race/Ethnicity

Gender	Race/Hispanic origin w/ NH Asian	Avg_Systolic_BP	Avg_Diastolic_BP	StdDev_Systolic_BP	StdDev_Diastolic_BP
1	1	125.814	74.48547	16.87686	10.90135
1	2	126.5696	75.79612	18.14925	11.69723
1	3	126.3709	74.67368	16.97677	11.40515
1	4	130.8836	77.98931	20.46936	13.12634
1	6	124.8698	75.82825	16.26046	10.52634
1	7	126.1311	76.26776	16.87232	12.0557
2	1	117.9475	72.09116	20.06057	10.84645
2	2	121.3669	72.97041	21.09274	11.50019
2	3	121.8841	73.19044	20.19042	11.17792
2	4	127.0132	77.84344	22.96148	12.88563
2	6	119.8122	73.5203	19.84175	10.80434
2	7	119	75.78947	18.82826	11.65772

Descriptive Statistics

Descriptive analysis provided an overview of blood pressure distributions and basic statistics (mean, standard deviation, etc.) across the entire population. Visual representations created using PROC SGPLOT highlighted variations in blood pressure across different demographic groups, including gender, race/ethnicity, and education levels.

Table 3: Frequency Distribution for Categorical Variables

Frequency Distribution for Categorical Variables

The FREQ Procedure

Gender				
RIAGENDR	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	3274	49.15	3274	49.15
2	3387	50.85	6661	100.00

Race/Hispanic origin w/ NH Asian				
RIDRETH3	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	706	10.60	706	10.60
2	647	9.71	1353	20.31
3	2469	37.07	3822	57.38
4	1749	26.26	5571	83.64
6	755	11.33	6326	94.97
7	335	5.03	6661	100.00

Six month time period				
RIDEXMON	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	3474	52.15	3474	52.15
2	3187	47.85	6661	100.00

Education level - Adults 20+				
DMDEDUC2	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	402	6.04	402	6.04
2	708	10.63	1110	16.66
3	1608	24.14	2718	40.80
4	2256	33.87	4974	74.67
5	1684	25.28	6658	99.95
7	1	0.02	6659	99.97
9	2	0.03	6661	100.00

Coded cuff size - oscillometric				
BPAOCSZ	Frequency	Percent	Cumulative Frequency	Cumulative Percent
2	29	0.44	29	0.44
3	2587	38.84	2616	39.27
4	3560	53.45	6176	92.72
5	485	7.28	6661	100.00

Table 4: Frequency Distribution for Categorized Continuous Variables

Frequency Distribution for Categorized Continuous Variables

The FREQ Procedure

age_group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
20-40	2115	31.75	2115	31.75
41-60	2293	34.42	4408	66.18
Over 60	2253	33.82	6661	100.00

sys_bp_group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
(120-140)	2334	35.04	2334	35.04
(141-200)	1277	19.17	3611	54.21
(>200)	18	0.27	3629	54.48
<120)	3032	45.52	6661	100.00

dia_bp_group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
(116-141)	19	0.29	19	0.29
(76-116)	3006	45.13	3025	45.41
(<76)	3636	54.59	6661	100.00

poverty_ratio_group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
(0-2)	3013	45.23	3013	45.23
(2-3)	1077	16.17	4090	61.40
(3-5)	2571	38.60	6661	100.00

Figure 1: Box Plot of Systolic Blood Pressure by Gender

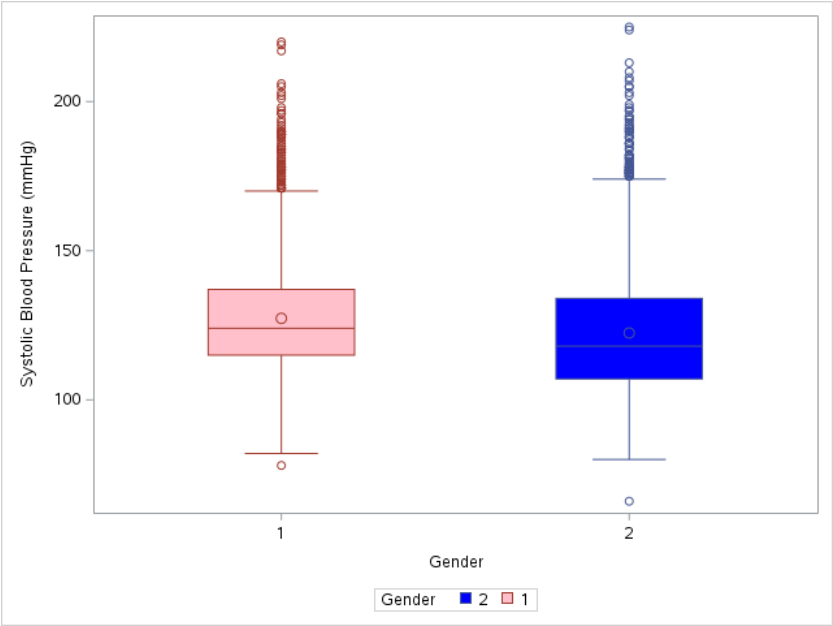


Figure 2: Box Plot of Diastolic Blood Pressure by Gender

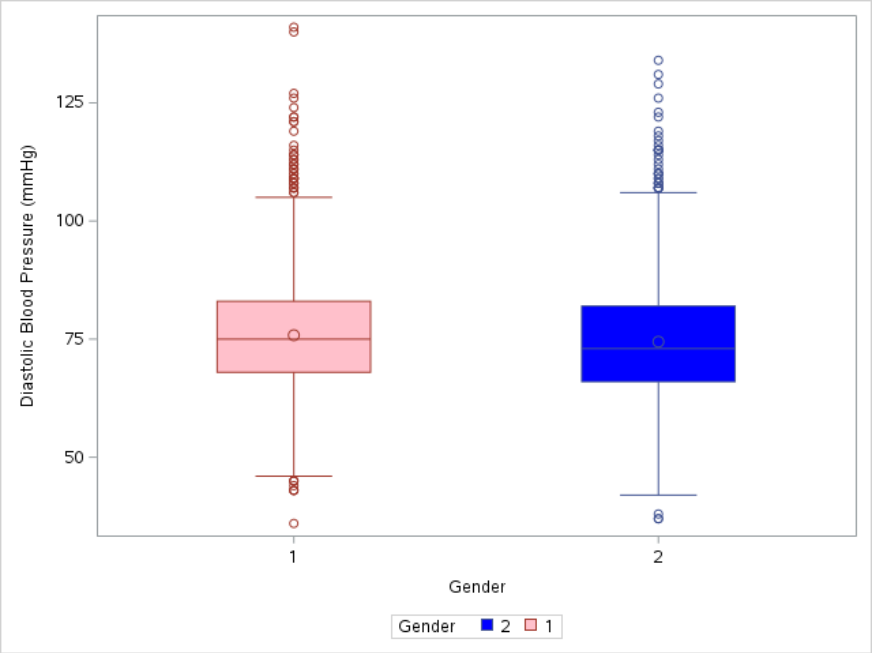


Figure 3: Distribution of BPXOSY1

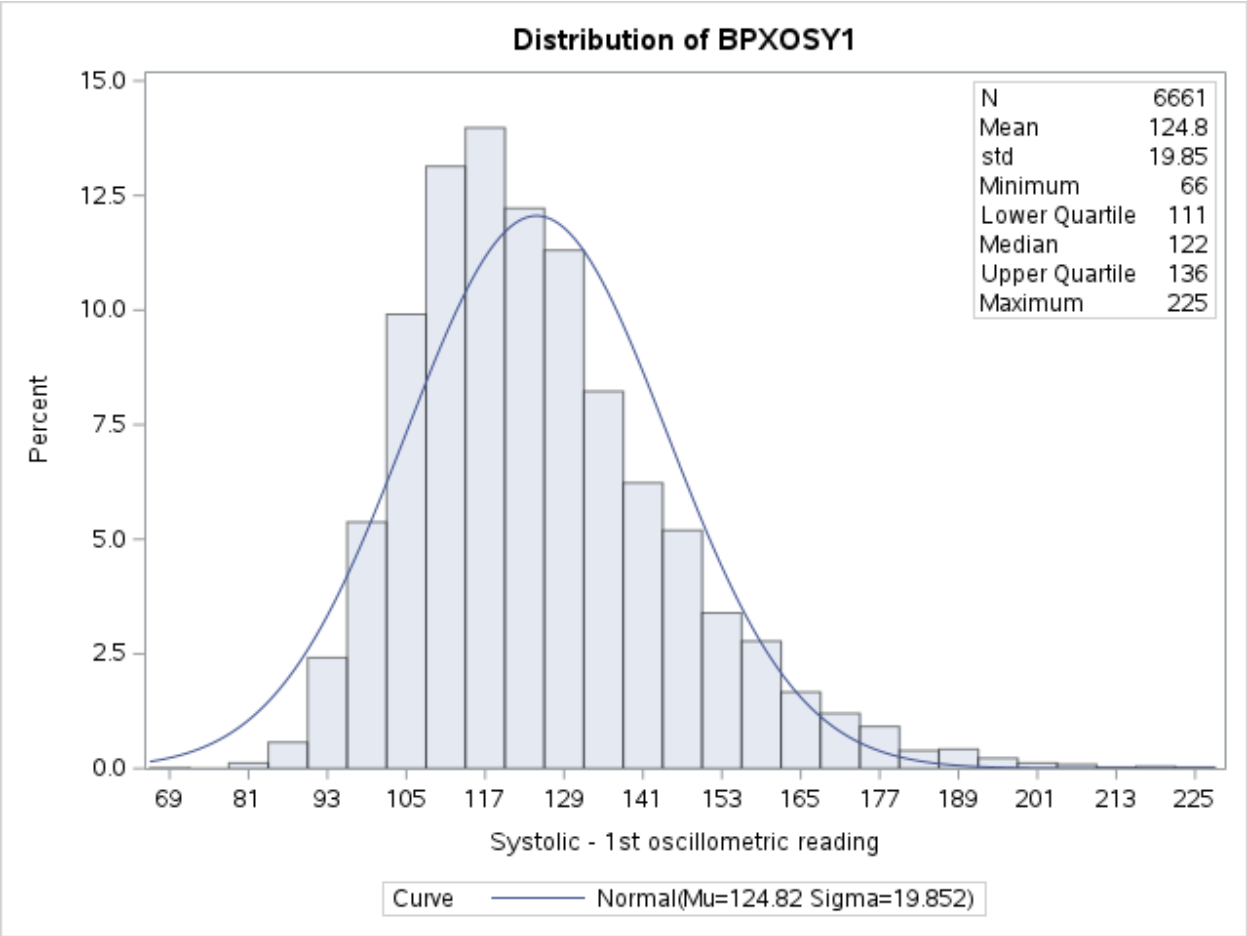


Figure 4: Distribution of BPXOSY1

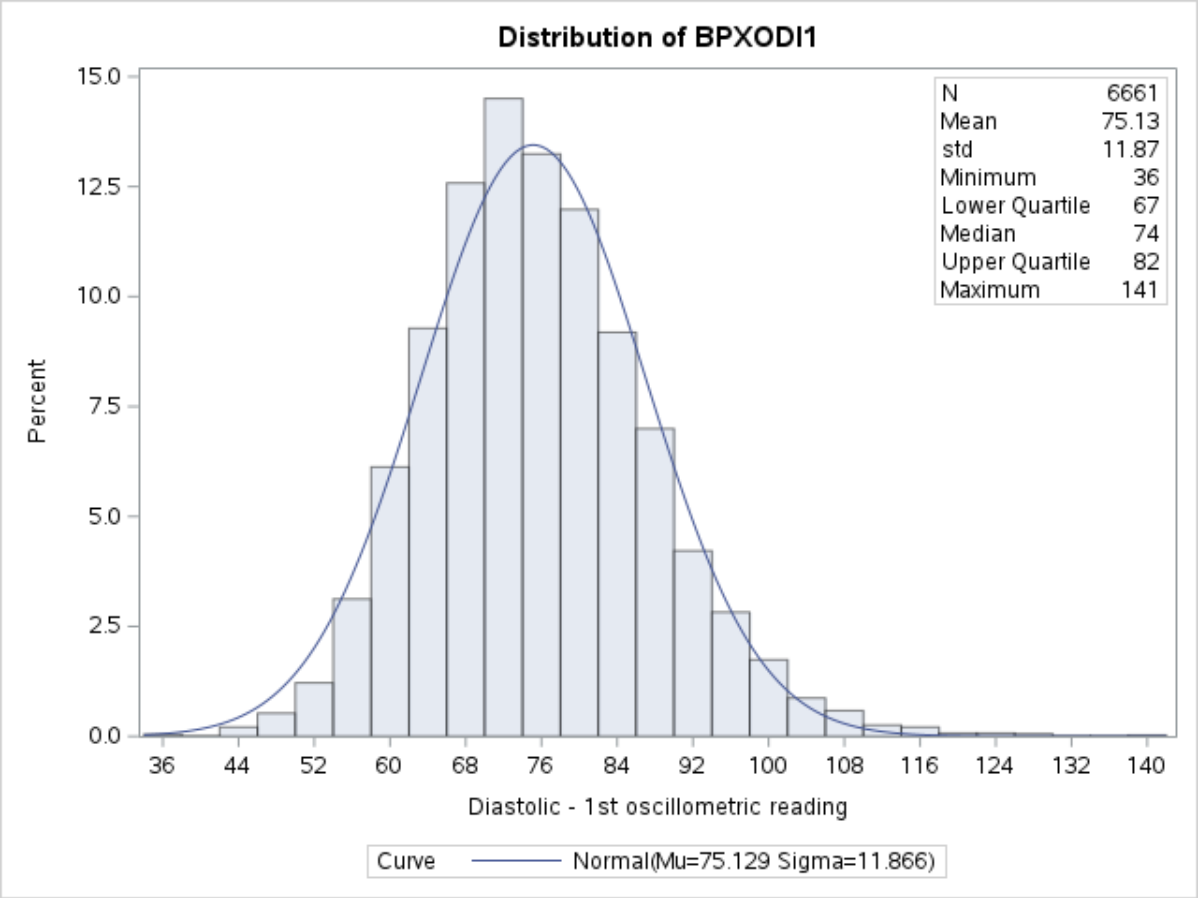


Table 5: Correlation Matrix for Continuous Variables

Correlation Matrix for Continuous Variables

The CORR Procedure

2 With Variables:	RIDAGEYR INDFMPIR
4 Variables:	RIDAGEYR INDFMPIR BPXOSY1 BPXODI1

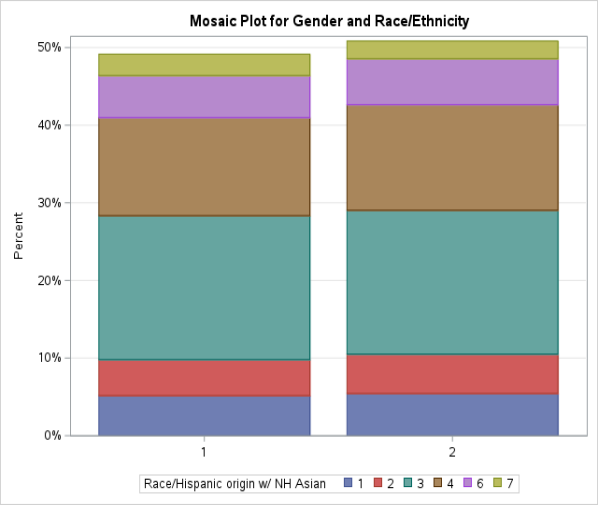
Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum	Label
RIDAGEYR	6661	50.85257	17.36362	338729	20.00000	80.00000	Age in years at screening
INDFMPIR	6661	2.61373	1.62772	17410	0	5.00000	Ratio of family income to poverty
BPXOSY1	6661	124.82225	19.85165	831441	66.00000	225.00000	Systolic - 1st oscillometric reading
BPXODI1	6661	75.12881	11.86608	500433	36.00000	141.00000	Diastolic - 1st oscillometric reading

Pearson Correlation Coefficients, N = 6661
Prob > |r| under H0: Rho=0

	RIDAGEYR	INDFMPIR	BPXOSY1	BPXODI1
RIDAGEYR Age in years at screening	1.00000	0.06904 <.0001	0.43900 <.0001	0.02562 0.0366
INDFMPIR Ratio of family income to poverty	0.06904 <.0001	1.00000	-0.04717 0.0001	-0.02003 0.1022

Figure 4: Mosaic Plot for Gender and Race/Ethnicity and for Gender and Education



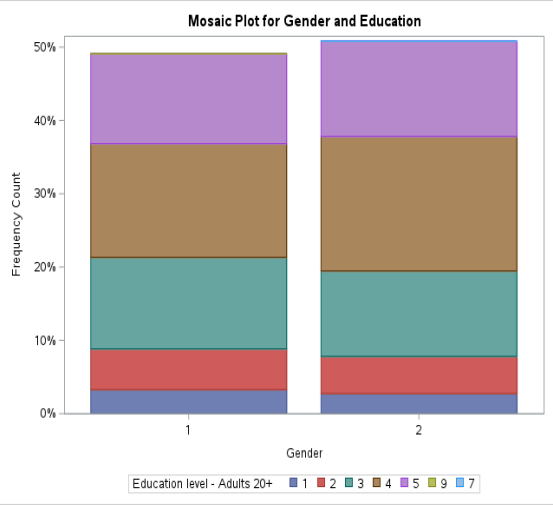
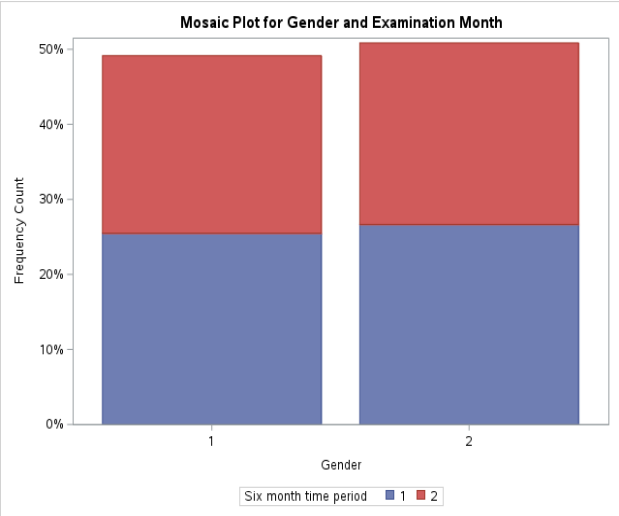


Figure 4: Mosaic Plot for Gender and Examination Month and for Race/Ethnicity and Education



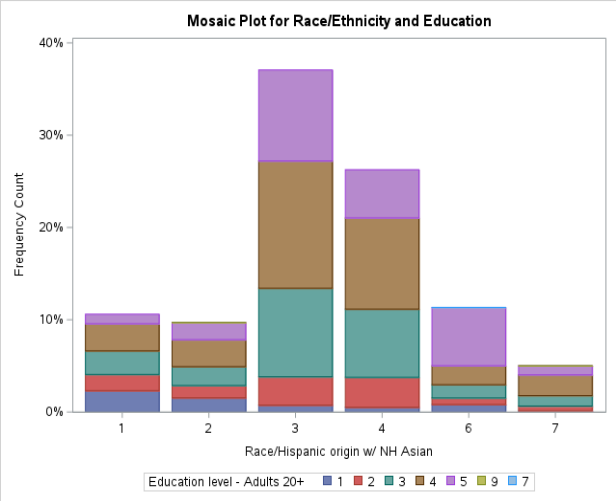
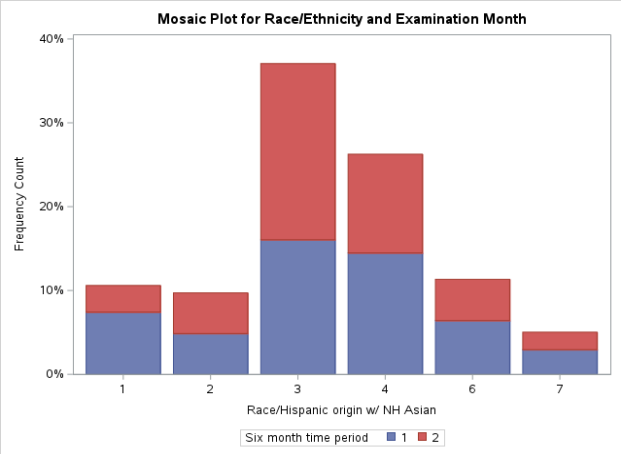


Figure 4: Mosaic Plot for Race/Ethnicity and Examination Month and for Education and Examination Month



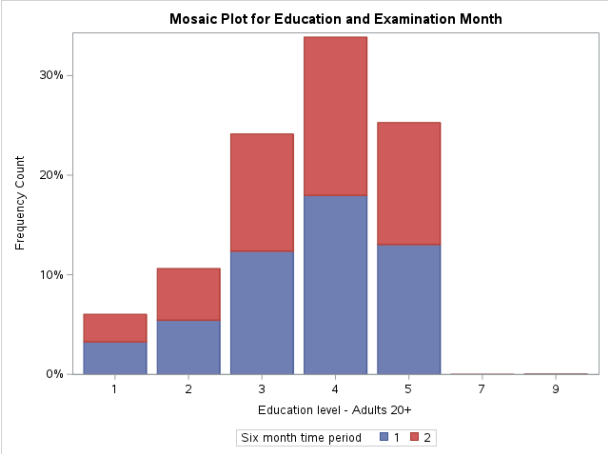
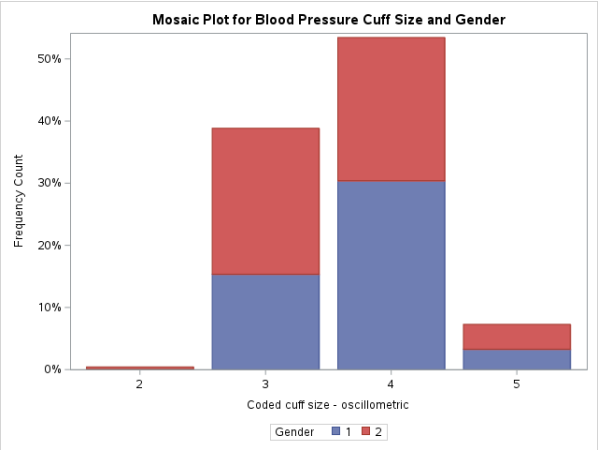


Figure 4: Mosaic Plot for Blood Pressure Cuff Size and Gender



Inferential Statistics

Statistical tests such as the two-sample t-test, chi-square tests, regression analysis, and correlation were conducted to examine the relationships between blood pressure and demographic variables. For instance, regression models assessed how age, gender, and socioeconomic status influence blood pressure outcomes. The results indicated significant differences in blood pressure levels across different age groups and socioeconomic statuses, suggesting targeted interventions could be beneficial.

Model

The GENMOD Procedure

Model Information		
Data Set	WORK.COMPLETE_DATA	
Distribution	Normal	
Link Function	Identity	
Dependent Variable	BPXOSY1	Systolic - 1st oscillometric reading

Number of Observations Read	6661
Number of Observations Used	6661

Class Level Information		
Class	Levels	Values
RIAGENDR	2	1 2
RIDEXMON	2	1 2
RIDRETH3	6	1 2 3 4 6 7
DMDEDUC2	7	1 2 3 4 5 7 9
BPAOCSZ	4	2 3 4 5

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	6642	2009301.3219	302.5145
Scaled Deviance	6642	6661.0000	1.0029
Pearson Chi-Square	6642	2009301.3219	302.5145
Scaled Pearson X2	6642	6661.0000	1.0029
Log Likelihood		-28466.2824	
Full Log Likelihood		-28466.2824	
AIC (smaller is better)		56972.5647	
AICC (smaller is better)		56972.6912	
BIC (smaller is better)		57108.6452	

Table 6: Analysis of Maximum Likelihood Parameter Estimates

Analysis Of Maximum Likelihood Parameter Estimates								
Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	90.6530	12.3613	66.4253	114.8807	53.78	<.0001
RIDAGEYR		1	0.4986	0.0127	0.4737	0.5235	1540.40	<.0001
RIAGENDR	1	1	4.4404	0.4345	3.5888	5.2920	104.44	<.0001
RIAGENDR	2	0	0.0000	0.0000	0.0000	0.0000	.	.
RIDEXMON	1	1	0.1643	0.4325	-0.6834	1.0121	0.14	0.7040
RIDEXMON	2	0	0.0000	0.0000	0.0000	0.0000	.	.
RIDRETH3	1	1	-1.4463	1.1735	-3.7463	0.8537	1.52	0.2178
RIDRETH3	2	1	-0.8701	1.1783	-3.1796	1.4394	0.55	0.4603
RIDRETH3	3	1	-1.9435	1.0229	-3.9484	0.0615	3.61	0.0575
RIDRETH3	4	1	4.7321	1.0394	2.6948	6.7693	20.73	<.0001
RIDRETH3	6	1	-0.1985	1.1731	-2.4978	2.1008	0.03	0.8656
RIDRETH3	7	0	0.0000	0.0000	0.0000	0.0000	.	.
DMDEDUC2	1	1	8.1300	12.3273	-16.0310	32.2911	0.43	0.5096
DMDEDUC2	2	1	6.5123	12.3152	-17.6251	30.6497	0.28	0.5969
DMDEDUC2	3	1	5.1756	12.3070	-18.9458	29.2969	0.18	0.6741
DMDEDUC2	4	1	4.9543	12.3058	-19.1647	29.0733	0.16	0.6872
DMDEDUC2	5	1	2.5747	12.3113	-21.5550	26.7045	0.04	0.8343
DMDEDUC2	7	1	-3.5314	21.2982	-45.2751	38.2123	0.03	0.8683
DMDEDUC2	9	0	0.0000	0.0000	0.0000	0.0000	.	.
INDFMPIR		1	-0.3852	0.1525	-0.6842	-0.0862	6.38	0.0116
BPAOCSZ	2	1	-5.0149	3.3315	-11.5445	1.5147	2.27	0.1322
BPAOCSZ	3	1	3.3917	0.8796	1.6677	5.1157	14.87	0.0001
BPAOCSZ	4	1	2.3126	0.8500	0.6467	3.9786	7.40	0.0065
BPAOCSZ	5	0	0.0000	0.0000	0.0000	0.0000	.	.
Scale		1	17.3681	0.1505	17.0757	17.6656		

Next one with BPXODI1

The GENMOD Procedure

Model Information		
Data Set	WORK.COMPLETE_DATA	
Distribution	Normal	
Link Function	Identity	
Dependent Variable	BPXODI1	Diastolic - 1st oscillometric reading

Number of Observations Read	6661
Number of Observations Used	6661

Class Level Information		
Class	Levels	Values
RIAGENDR	2	1 2
RIDEXMON	2	1 2
RIDRETH3	6	1 2 3 4 6 7
DMDEDUC2	7	1 2 3 4 5 7 9
BPAOCSZ	4	2 3 4 5

Criteria For Assessing Goodness Of Fit			
Criterion	DF	Value	Value/DF
Deviance	6642	887578.2610	133.6312
Scaled Deviance	6642	6661.0000	1.0029
Pearson Chi-Square	6642	887578.2610	133.6312
Scaled Pearson X2	6642	6661.0000	1.0029
Log Likelihood		-25745.1118	
Full Log Likelihood		-25745.1118	
AIC (smaller is better)		51530.2237	
AICC (smaller is better)		51530.3502	
BIC (smaller is better)		51666.3042	

Table 7: Analysis of Maximum Likelihood Parameter Estimates

Analysis Of Maximum Likelihood Parameter Estimates								
Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	75.3100	8.2157	59.2075	91.4125	84.03	<.0001
RIDAGEYR		1	0.0284	0.0084	0.0119	0.0450	11.35	0.0008
RIAGENDR	1	1	0.8120	0.2888	0.2460	1.3780	7.91	0.0049
RIAGENDR	2	0	0.0000	0.0000	0.0000	0.0000	.	.
RIDEXMON	1	1	0.2290	0.2875	-0.3344	0.7925	0.63	0.4257
RIDEXMON	2	0	0.0000	0.0000	0.0000	0.0000	.	.
RIDRETH3	1	1	-2.7344	0.7799	-4.2630	-1.2057	12.29	0.0005
RIDRETH3	2	1	-1.5046	0.7832	-3.0396	0.0303	3.69	0.0547
RIDRETH3	3	1	-1.7513	0.6799	-3.0838	-0.4187	6.63	0.0100
RIDRETH3	4	1	1.8256	0.6908	0.4715	3.1796	6.98	0.0082
RIDRETH3	6	1	0.3656	0.7797	-1.1626	1.8937	0.22	0.6391
RIDRETH3	7	0	0.0000	0.0000	0.0000	0.0000	.	.
DMDEDUC2	1	1	2.4610	8.1931	-13.5972	18.5192	0.09	0.7639
DMDEDUC2	2	1	1.2426	8.1851	-14.7998	17.2851	0.02	0.8793
DMDEDUC2	3	1	1.5611	8.1796	-14.4707	17.5928	0.04	0.8486
DMDEDUC2	4	1	1.7956	8.1788	-14.2346	17.8258	0.05	0.8262
DMDEDUC2	5	1	0.9161	8.1825	-15.1213	16.9535	0.01	0.9109
DMDEDUC2	7	1	-1.6971	14.1554	-29.4413	26.0470	0.01	0.9046
DMDEDUC2	9	0	0.0000	0.0000	0.0000	0.0000	.	.
INDFMPIR		1	-0.0395	0.1014	-0.2382	0.1592	0.15	0.6968
BPAOCSZ	2	1	-9.8135	2.2142	-14.1532	-5.4737	19.64	<.0001
BPAOCSZ	3	1	-5.3186	0.5846	-6.4645	-4.1728	82.76	<.0001
BPAOCSZ	4	1	-1.6228	0.5649	-2.7301	-0.5156	8.25	0.0041
BPAOCSZ	5	0	0.0000	0.0000	0.0000	0.0000	.	.
Scale		1	11.5434	0.1000	11.3490	11.7411		

Concluding Remarks

This study highlighted significant demographic determinants of blood pressure among adults. Age and socioeconomic factors play a crucial role in influencing blood pressure levels. The insights gained emphasize the importance of statistical analysis in understanding complex health issues and can inform public health strategies aimed at reducing the prevalence of hypertension.

References

National Center for Health Statistics. (2018). National Health and Nutrition Examination Survey Data. Hyattsville, MD: Centers for Disease Control and Prevention. Retrieved from https://www.cdc.gov/nchs/nhanes/index.htm

SAS Institute Inc. (2018). SAS/STAT® 15.1 User's Guide. Cary, NC: SAS Institute Inc.

Cody, R. (2018). *Learning SAS by Example* (2nd ed.). SAS Institute Inc.. https://online.vitalsource.com/books/9781635266580

Wang, Y., & Wang, Q. J. (2004). The prevalence of prehypertension and hypertension among US adults according to the new joint national committee guidelines: new challenges of the old problem. *Archives of internal medicine*, *164*(19), 2126-2134.

Appendix

```
libname myxpt XPORT '/home/u63740333/Demo.XPT';
proc contents data=myxpt._all_ out=contents_details(keep=memname type) noprint;
run;

proc print data=contents_details;
run;
data demo;
   set myxpt.demo;
run;
proc print data=demo(obs=10);
run;
libname myxpt XPORT '/home/u63740333/Demo.XPT';

data demo;
   set myxpt.P_DEMO;
run;

proc print data=demo(obs=10);
run;
libname mybpx XPORT '/home/u63740333/BPXO.XPT';
proc contents data=mybpx._all_ out=bpx_contents_details(keep=memname type) noprint;
run;
proc print data=bpx_contents_details;
run;
libname mybpx XPORT '/home/u63740333/BPXO.XPT';

data bpx;
   set mybpx.P_BPXO;
run;
proc print data=bpx(obs=10);
run;

libname myxpt XPORT '/home/u63740333/Demo.XPT';
libname mybpx XPORT '/home/u63740333/BPXO.XPT';

/* Sort the 'demo' dataset by SEQN */
```

```
proc sort data=myxpt.P_DEMO out=demo_sorted;
   by SEQN;
run;

/* Sort the 'bpx' dataset by SEQN */
proc sort data=mybpx.P_BPXO out=bpx_sorted;
   by SEQN;
run;

/* Merge the two sorted datasets by SEQN with specific variables included */
data merged_data;
   merge demo_sorted (in=in_demo keep=SEQN RIAGENDR RIDAGEYR RIDRETH3 RIDEXMON
DMDEDUC2 INDFMPIR)
        bpx_sorted (in=in_bpx keep=SEQN BPAOCSZ BPXOSY1 BPXODI1);
   by SEQN;
   /* Include only observations that appear in both datasets */
   if in_demo and in_bpx;
run;

/* Print the first 10 observations of the refined merged dataset */
proc print data=merged_data(obs=10);
run;
proc sql;
  title "Average Blood Pressure by Gender and Race/Ethnicity";
  select RIAGENDR,
        RIDRETH3,
        mean(BPXOSY1) as Avg_Systolic_BP,
        mean(BPXODI1) as Avg_Diastolic_BP,
        std(BPXOSY1) as StdDev_Systolic_BP,
        std(BPXODI1) as StdDev_Diastolic_BP
  from complete_data
  group by RIAGENDR, RIDRETH3
  order by RIAGENDR, RIDRETH3;
quit;

proc freq data=merged_data;
   tables RIAGENDR RIDAGEYR RIDRETH3 RIDEXMON DMDEDUC2 INDFMPIR BPAOCSZ BPXOSY1
BPXODI1 / missing;
run;

proc sql;
   select count(*) as total_observations from merged_data;
```

```
quit;
data complete_data;
   set merged_data;
   if cmiss(SEQN, RIAGENDR, RIDAGEYR, RIDRETH3, RIDEXMON, DMDEDUC2, INDFMPIR,
BPAOCSZ, BPXOSY1, BPXODI1) = 0;
run;

proc print data=complete_data(obs=10);
run;
proc sql;
   select "Before removal" as Step, count(*) as Total_Observations from merged_data
   union
   select "After removal" as Step, count(*) as Total_Observations from complete_data;
run;

proc freq data=complete_data;
   tables RIAGENDR RIDRETH3 RIDEXMON DMDEDUC2 BPAOCSZ/ missing;
   title "Frequency Distribution for Categorical Variables";
run;

data categorized_data;
 length sys_bp_group $ 10;
 length dia_bp_group $ 10;
 length poverty_ratio_group  $ 10;
   set complete_data;

   /* Group Age into Ranges */
   if RIDAGEYR < 20 then age_group = 'Under 20';
   else if RIDAGEYR <= 40 then age_group = '20-40';
   else if RIDAGEYR <= 60 then age_group = '41-60';
   else if RIDAGEYR > 60 then age_group = 'Over 60';

   /* Group Systolic Blood Pressure into Ranges */
  if BPXOSY1 < 120 then sys_bp_group = '<120)';
   else if BPXOSY1 <= 140 then sys_bp_group = '(120-140)';
   else if BPXOSY1 <= 200 then sys_bp_group = '(141-200)';
   else if BPXOSY1 > 200 then sys_bp_group = '(>200)';

   /* Categorize Diastolic Blood Pressure (BPXODI1) */
   if BPXODI1 < 76 then dia_bp_group = '(<76)';
   else if BPXODI1 <= 116 then dia_bp_group = '(76-116)';
   else if BPXODI1 > 116 then dia_bp_group = '(116-141)';
```

```
   /* Categorize Income-to-Poverty Ratio (INDFMPIR) */
   if INDFMPIR < 2 then poverty_ratio_group = '(0-2)';
   else if INDFMPIR <= 3 then poverty_ratio_group = '(2-3)';
   else if INDFMPIR > 3 then poverty_ratio_group = '(3-5)';

run;

/* Box Plot for Systolic Blood Pressure by Gender with Color */
proc sgplot data=complete_data;
   vbox BPXOSY1 / category=RIAGENDR group=RIAGENDR;
   xaxis label="Gender";
   yaxis label="Systolic Blood Pressure (mmHg)";
   styleattrs datacolors=(blue pink); /* Customize colors for male and female, respectively */
run;

/* Box Plot for Diastolic Blood Pressure by Gender with Color */
proc sgplot data=complete_data;
   vbox BPXODI1 / category=RIAGENDR group=RIAGENDR;
   xaxis label="Gender";
   yaxis label="Diastolic Blood Pressure (mmHg)";
   styleattrs datacolors=(blue pink); /* Customize colors for male and female, respectively */
run;


proc freq data=categorized_data;
   tables age_group sys_bp_group dia_bp_group poverty_ratio_group/ missing;
   title "Frequency Distribution for Categorized Continuous Variables";
run;
/* Output from PROC UNIVARIATE */

proc means data=complete_data N mean std min p25 median p75 max;
   var  BPXOSY1 BPXODI1;
   title "Summary Statistics for Continuous Variables";
run;

proc univariate data=complete_data;
   var BPXOSY1 BPXODI1;
   histogram / normal;
   inset n mean (5.2) std='std' (5.2) min p25 median p75 max / position=ne;
   title "Detailed Analysis of Blood Pressure Variables";
```

```
run;


proc corr data=complete_data;
   var RIDAGEYR RIAGENDR RIDRETH3 RIDEXMON DMDEDUC2 INDFMPIR;
   with RIDAGEYR RIAGENDR RIDRETH3 RIDEXMON DMDEDUC2 INDFMPIR;
run;
proc freq data=complete_data;
   tables RIAGENDR*RIDRETH3 / chisq;
   tables RIAGENDR*DMDEDUC2 / chisq;
   tables RIAGENDR*RIDEXMON / chisq;
   tables RIDRETH3*DMDEDUC2 / chisq;
   tables RIDRETH3*RIDEXMON / chisq;
   tables DMDEDUC2*RIDEXMON / chisq;
   tables BPAOCSZ*RIAGENDR / chisq;
   title "Chi-Square Tests for Independence Among Categorical Variables";
run;

/* First, use PROC FREQ to create a dataset containing the counts for plotting */
proc freq data=complete_data noprint;
   tables RIAGENDR*RIDRETH3 / out=RIAGENDR_RIDRETH3_list;
   tables RIAGENDR*DMDEDUC2 / out=RIAGENDR_DMDEDUC2_list;
   tables RIAGENDR*RIDEXMON / out=RIAGENDR_RIDEXMON_list;
   tables RIDRETH3*DMDEDUC2 / out=RIDRETH3_DMDEDUC2_list;
   tables RIDRETH3*RIDEXMON / out=RIDRETH3_RIDEXMON_list;
   tables DMDEDUC2*RIDEXMON / out=DMDEDUC2_RIDEXMON_list;
   tables BPAOCSZ*RIAGENDR / out=BPAOCSZ_RIAGENDR_list;
run;

/* Mosaic Plot for RIAGENDR and RIDRETH3 */
proc sgplot data=RIAGENDR_RIDRETH3_list;
   vbar RIAGENDR / response=count group=RIDRETH3 stat=percent;
   xaxis display=(nolabel);
   yaxis grid label="Percent";
   title "Mosaic Plot for Gender and Race/Ethnicity";
run;

/* Repeat for each variable pair */
proc sgplot data=RIAGENDR_DMDEDUC2_list;
   vbar RIAGENDR / response=count group=DMDEDUC2 stat=percent;
   title "Mosaic Plot for Gender and Education";
run;
```

```
proc sgplot data=RIAGENDR_RIDEXMON_list;
   vbar RIAGENDR / response=count group=RIDEXMON stat=percent;
   title "Mosaic Plot for Gender and Examination Month";
run;

proc sgplot data=RIDRETH3_DMDEDUC2_list;
   vbar RIDRETH3 / response=count group=DMDEDUC2 stat=percent;
   title "Mosaic Plot for Race/Ethnicity and Education";
run;

proc sgplot data=RIDRETH3_RIDEXMON_list;
   vbar RIDRETH3 / response=count group=RIDEXMON stat=percent;
   title "Mosaic Plot for Race/Ethnicity and Examination Month";
run;

proc sgplot data=DMDEDUC2_RIDEXMON_list;
   vbar DMDEDUC2 / response=count group=RIDEXMON stat=percent;
   title "Mosaic Plot for Education and Examination Month";
run;

proc sgplot data=BPAOCSZ_RIAGENDR_list;
   vbar BPAOCSZ / response=count group=RIAGENDR stat=percent;
   title "Mosaic Plot for Blood Pressure Cuff Size and Gender";
run;


proc reg data=complete_data;
   model BPXOSY1 = RIDAGEYR RIAGENDR RIDRETH3 RIDEXMON DMDEDUC2 INDFMPIR
BPAOCSZ/ vif;
run;

proc reg data=complete_data;
   model BPXODI1 = RIDAGEYR RIAGENDR RIDRETH3 RIDEXMON DMDEDUC2 INDFMPIR BPAOCSZ
/ vif;
run;
proc genmod data=complete_data;
   class RIAGENDR RIDEXMON RIDRETH3 DMDEDUC2 BPAOCSZ;
   model BPXOSY1 = RIDAGEYR RIAGENDR RIDEXMON RIDRETH3 DMDEDUC2 INDFMPIR
BPAOCSZ / dist=normal link=identity;
run;
proc genmod data=complete_data;
```

```
   class RIAGENDR RIDEXMON RIDRETH3 DMDEDUC2 BPAOCSZ;
   model BPXODI1 = RIDAGEYR RIAGENDR RIDEXMON RIDRETH3 DMDEDUC2 INDFMPIR BPAOCSZ
/ dist=normal link=identity;
run;



proc ttest data=complete_data;
   class RIAGENDR; /* Replace with other categorical variables as needed */
   var BPXOSY1 BPXODI1;
   title "T-test for Blood Pressure by Gender";
run;

proc ttest data=complete_data;
   class  RIDEXMON ; /* Replace with other categorical variables as needed */
   var BPXOSY1 BPXODI1;
   title "T-test for Blood Pressure by Gender";
run;

proc ttest data=complete_data;
   class RIDRETH3; /* Replace with other categorical variables as needed */
   var BPXOSY1 BPXODI1;
   title "T-test for Blood Pressure by Gender";
run;
proc ttest data=complete_data;
   class DMDEDUC2; /* Replace with other categorical variables as needed */
   var BPXOSY1 BPXODI1;
   title "T-test for Blood Pressure by Gender";
run;
```